Microsoft Elevate

CAPSTONE PROJECT

SUPERSTORE SALES PERFORMANCE DASHBOARD

PRESENTED BY

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OUTLINE:

- **Problem Statement** (Should not include solution)
- **Proposed System/Solution**
- **System Development Approach** (Technology Used)
- **Algorithm & Deployment**
- **Result (Output Image)**
- **Conclusion**
- **Future Scope**
- **References**

PROBLEM STATEMENT:

Retail businesses generate large volumes of sales data from different regions, product categories, and customer segments. Analyzing this data manually becomes difficult and time-consuming.

Organizations need an efficient system to:

- Monitor overall sales performance
- Identify profitable and loss-making products
- Analyze sales trends across months and regions
- Understand customer segments and shipping preferences
- Support data-driven business decision making
- However, raw sales data in spreadsheets does not provide clear insights.
- Therefore, a **data visualization dashboard is required to transform raw sales data into meaningful and interactive business insights.**

PROPOSED SOLUTION:

To address the problem of analyzing large volumes of sales data, a **Power BI based Sales Performance Dashboard** is developed. The proposed system provides an interactive data visualization platform that helps analyze and monitor sales performance effectively.

The solution includes the following components:

Data Collection

• Import Superstore sales dataset from Excel.

Data Preprocessing

• Clean and transform data using **Power Query**.

• Handle missing or inconsistent values.

Data Analysis

• Use Power BI to analyze sales, profit, quantity, and order information.

Dashboard Visualization

Create interactive visualizations such as:

- Sales by Region
- Sales by Category
- Sales by Customer Segment
- Monthly Sales Trend

Interactive Filters

Users can filter data by:

- Year
- Quarter
- Month
- Category

This dashboard helps businesses gain **clear insights and make data-driven decisions**.

SYSTEM APPROACH:

This project follows a data visualization approach using Power BI to analyze the Superstore sales dataset and generate meaningful business insights.

System Requirements

Hardware

- Computer/Laptop
- Minimum 8 GB RAM

Software

- Microsoft Power BI Desktop
- Microsoft Excel

Technologies Used

- Power BI
- Power Query (Data Transformation)
- DAX – Data Analysis Expressions

Dataset Used

Superstore Sales Dataset containing:

- Order ID
- Sales
- Profit
- Quantity
- Category
- Region
- Customer Segment

This approach helps transform raw sales data into **interactive dashboards for better analysis and decision making**.

ALGORITHM & DEPLOYMENT:

Data Input

•The Superstore sales dataset is imported into **Power BI** from an Excel file.

•The dataset contains information such as **Order ID, Sales, Profit, Quantity, Category, Region, and Customer Segment**.

Data Cleaning & Transformation

•Data preprocessing is performed using **Power Query**.

•Missing values and inconsistencies are handled.

•Data fields are formatted for analysis.

Data Modeling

•Relationships between data fields are created.

•Measures and calculated columns are defined using **DAX (Data Analysis Expressions)**.

Visualization Process

•Interactive charts and graphs are created to analyze:

- Sales performance
- Profit trends
- Regional distribution
- Customer segments

Deployment

•The dashboard is published through **Power BI Service** for sharing and real-time access.

RESULT:

The Power BI dashboard successfully visualizes the Superstore sales dataset and provides meaningful insights into business performance.

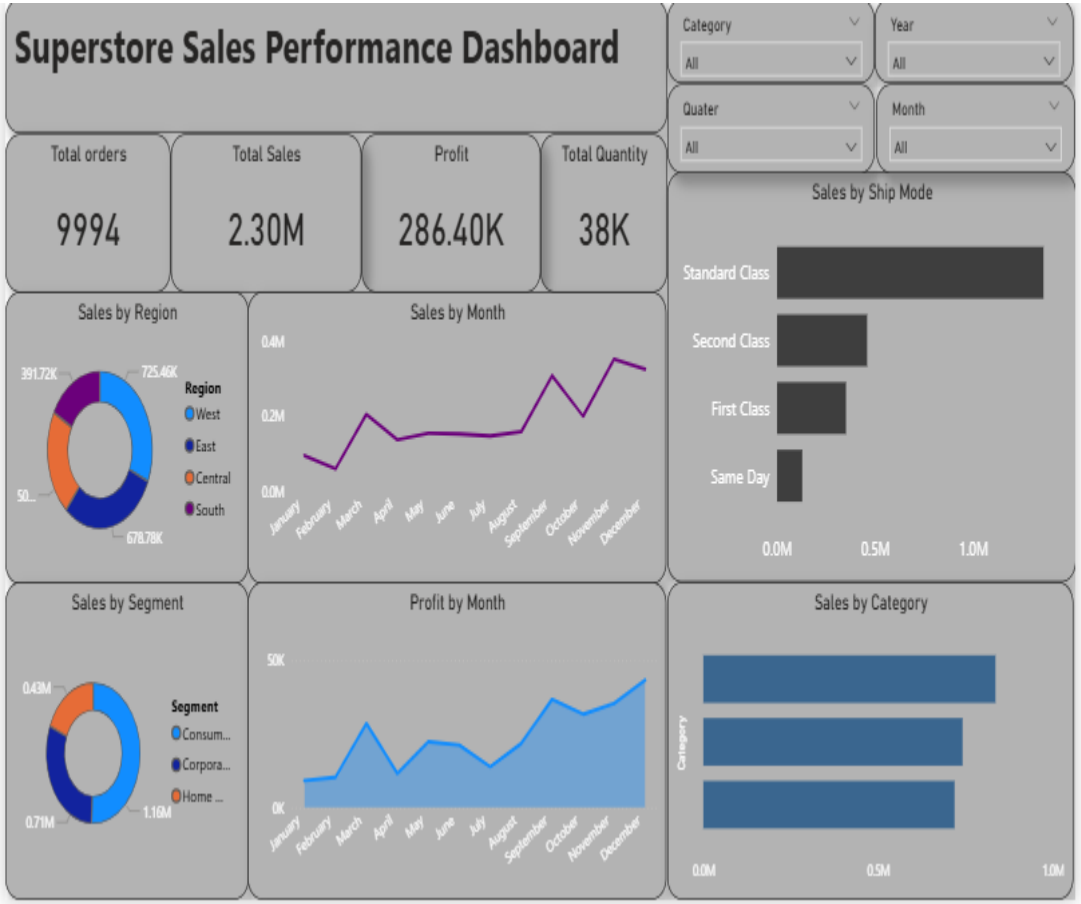
The dashboard displays key performance indicators such as:

- Total Sales
- Total Profit
- Total Orders
- Total Quantity Sold

It also provides detailed analysis through visualizations such as:

- Sales by Region
- Monthly Sales Trend
- Profit by Month
- Sales by Category
- Sales by Customer Segment
- Shipping Mode Analysis

The interactive filters allow users to analyze data based on **year, quarter, month, and category**, enabling better decision-making.



CONCLUSION:

- The Power BI dashboard successfully analyzes the Superstore sales dataset and provides clear insights into business performance.
- The dashboard highlights important metrics such as **total sales, profit, orders, and quantity sold**. It also visualizes sales distribution by **region, category, customer segment, and shipping mode**.
- The interactive filters enable users to explore data across **different years, quarters, and months**, making the analysis more flexible and informative.
- Overall, the dashboard helps businesses **understand sales trends, identify profitable areas, and support data-driven decision making.**

FUTURE SCOPE:

The project can be further enhanced with additional features to improve data analysis and decision-making.

- Integrate **real-time sales data** for live business monitoring
- Add **predictive analytics using machine learning** to forecast future sales trends
- Expand the dashboard with **customer behavior and product performance analysis**
- Deploy the dashboard on **Power BI Service or cloud platforms** for easy access and sharing
- Include **advanced visualizations and automated reporting systems**

These improvements can help organizations gain **deeper insights and better strategic planning**.

REFERENCES:

The development of this project was supported by various learning resources and documentation. The primary reference used was the official **Microsoft Power BI Documentation and Microsoft Learn platform**, which provides detailed guidance on Power BI tools and features. The **Superstore Sales Dataset obtained from Kaggle** was used for performing the data analysis and visualization. Additional knowledge and best practices for data visualization were gathered from **Power BI tutorials available on the Microsoft website and Power BI community resources**.

GitHub Link: https://github.com/Uday2kranth/Superstore

Thank You